

Spring AI: Module 10

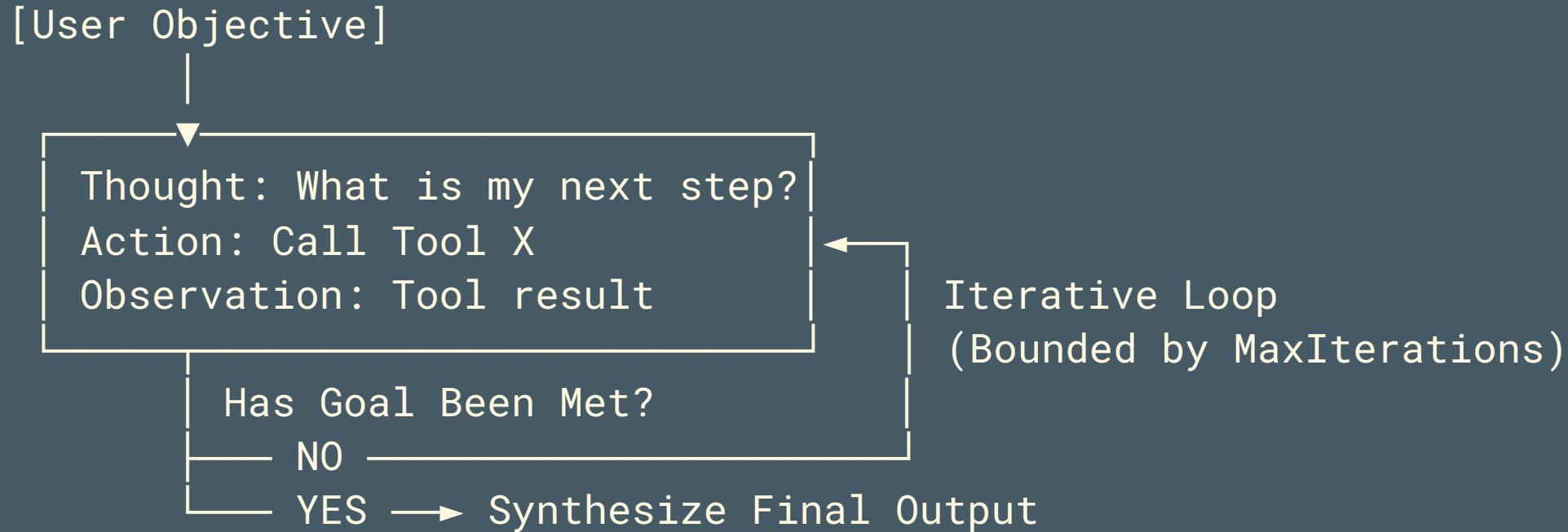
Agentic AI & Autonomous Multi-Agent Workflows

Reasoning, Acting & Squad Delegation in Java

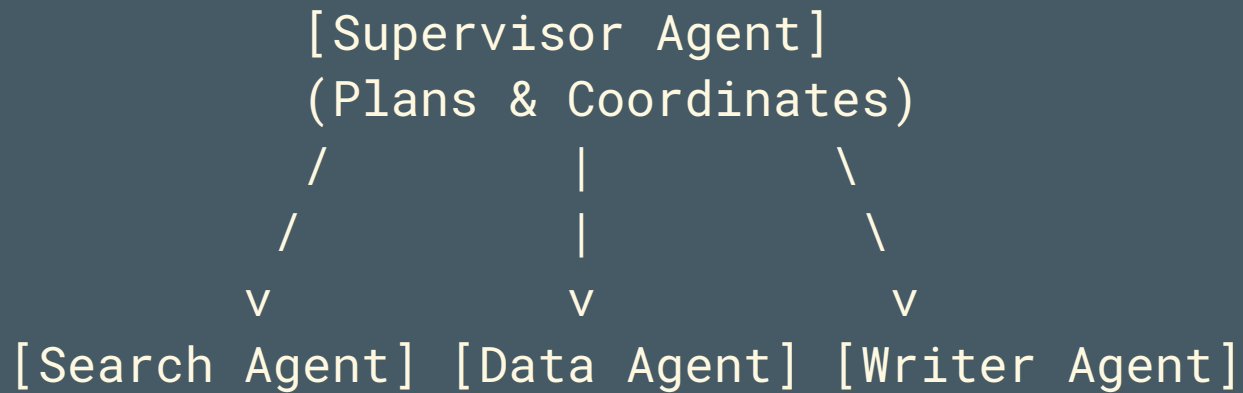
1. What Makes an LLM "Agentic"?

- Standard LLMs generate a single completion for a single prompt.
- **Agentic AI** autonomously:
 1. Breaks down an ambiguous objective into sub-goals.
 2. Executes tools in a feedback loop (Thought -> Action -> Observation).
 3. Adapts its plan based on tool results until the goal is solved.

2. The ReAct Pattern in Spring AI



3. Multi-Agent Squads: Supervisor-Worker Architecture



- Each worker agent has isolated prompt instructions, specific tools, and bounded execution scopes.

4. Operational Safety: Loop Detection & Human Approval

- Without safety bounds, agents can enter infinite reasoning loops.
- **Critical Controls:**
 - Maximum iteration cutoff (e.g. 5 steps).
 - Loop detection (flagging identical repeated actions).
 - Human-in-the-loop checkpoints before destructive operations (e.g. database updates, financial transfers).

5. Key Architecture Takeaways

1. Always set strict iteration and timeout boundaries on agent loops.
2. Delegate specialized tasks to focused subagents rather than building one monolithic prompt.
3. Gate sensitive external mutations behind human approval checks.